D–6899

Sub. Code
31511

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

First Semester

Computer Application

DIGITAL COMPUTER ORGANIZATION

(CBCS 2018-19 Academic Year onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Subtract -72 from -50 using 2's complement method.

2. Simplify Y=(A+B)(A+C')(B'+C').

3. Define the term 'Half adder'.

4. State the basic operation of a decoder.

5. List the four phases of instruction cycle.

6. Mention the purpose of accumulator and program counter.

7. What is the need for interface between I/O device and CPU?

8. Define the term peripherals. Mention its types.

9. What is the purpose of Main memory?

10. Write the uses of virtual memory.

PART B — (5 × 5 = 25 marks)

Answer ALL questions, Choosing either (a) or (b).

11. (a) Describe the fundamentals of Boolean algebra.

Or

(b) Write short notes on Quine-Mckluskey method.

12. (a) Perform D to T flip-flop conversion.

Or

(b) Describe briefly about BCD counter.

13. (a) Brief on computer registers.

Or

(b) Draw and explain the design of basic computer.

14. (a) Write short notes on stack organization.

Or

(b) What is Direct Memory Access? Explain briefly.

15. (a) Discuss about associative memory.

Or

(b) Brief on memory management hardware.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Simplify the following Boolean function using K–map F $F(w, x, y, z) = \sum m(0,1,2,4,5,6,8,9,12,13,14)$.

17. With neat block diagram, explain the operation of shift register.

18. Detail on different addressing modes.

19. Explain in detail about various addressing modes.

20. Discuss on cache mapping techniques.

D–6900

Sub. Code
31512

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

First Semester

OBJECT ORIENTED PROGRAMMING AND C++

(CBCS 2018-19 Aca. Year onwards & 2020-21 Aca.Year onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is the need for streams?

2. What is manipulator? Write two manipulators in C++.

3. Define the terms class and object.

4. What is 'this' pointer?

5. How will you overload Unary and Binary operators using Friend function?

6. What is an abstract class?

7. Draw the hierarchy of file stream classes.

8. List any two ios functions.

9. What is meant by Exception?

10. Write the functions that handle uncaught exceptions.

PART B — (5 × 5 = 25 marks)

Answer ALL questions, Choosing either (a) or (b).

All questions carry equal marks.

11. (a) Briefly explain about object oriented programming paradigm.

Or

(b) Write short notes on formatted I/O.

12. (a) What do you mean by inline function? Explain with example code.

Or

(b) Explain copy constructor with suitable example.

13. (a) Describe Pure Virtual function with an example.

Or

(b) What is function overloading? Give an example.

14. (a) Explain with example how can a class template be created.

Or

(b) What is file? Write an example program for sequential access.

15. (a) Write short notes on catching exception. Give an example.

Or

(b) Write a program to throw exception from overloaded operator function.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

All questions carry equal marks.

16. Describe the basic concepts of Object Oriented Programming.

17. Describe the concept of call by reference & return by reference with program code.

18. Define the term Inheritance. Explain multilevel and hierarchal inheritance with program code.

19. Explain the concept of function template with examples.

20. Explain the use of try, catch and throw for exception handling in C++ through examples.

———————

D–6901

Sub. Code
31513

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

First Semester

Computer Application

DATA STRUCTURE AND ALGORITHMS

(CBCS 2018-19 Aca.Year onwards & 2020-21 Aca.year onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define the term data structure?

2. Define and write the characteristics of Algorithm.

3. What is a circular queue? Write the conditions of circular queue.

4. Differentiate between Array and Linked list.

5. What are ancestors and descendants?

6. What do you mean by the term Strictly Binary Tree?

7. State the applications of linear and binary search techniques.

8. What is the time complexity of binary search?

9. What is the best case time complexity of quick sort?

10. What is meant by external sorting?

PART B — (5 × 5 = 25 marks)

Answer ALL questions, Choosing either (a) or (b).

All questions carry equal marks.

11. (a) Explain briefly about various types of data structure.

Or

(b) How do you find the complexity of an algorithm? What is the relation between the time and space complexities of an algorithm? Justify your answer with an example.

12. (a) Explain the various applications of stack.

Or

(b) Convert the following Infix Expression to postfix using stack. A*B-(C+D)+E

13. (a) Explain the insertion and deletion operations of binary tree with example.

Or

(b) Write a recursive algorithm for binary tree traversal with an example.

14. (a) Compare working principle of binary search and linear search technique with example.

Or

(b) Write a program to search a number within a given set of numbers using binary search.

15. (a) Explain tree sort technique with an example.

Or

(b) Explain selection sort with illustration.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

All questions carry equal marks.

16. Define the term array. How are two-dimensional arrays represented in memory? Explain how address of an element is calculated in a two dimensional array.

17. Write an algorithm to perform the following operation on a doubly linked list.

(a) Insert new node at the beginning of the list.

(b) Insert new node at Middle.

(c) Delete a node at middle and at last.

(d) Count the number of nodes.

18. What is Hashing? Explain Different Hash function methods in detail.

19. Write short note on :

(a) Linear search

(b) Binary search.

20. Discuss on Bubble Sort with the help of an example.

D–6902

Sub. Code
31521

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Second Semester

Computer Applications

SOFTWARE ENGINEERING

(CBCS 2018-19 Academic Year onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Define software process.
2. Write down the measures, metrics and indicators of software process.
3. What do you mean by requirement elicitation?
4. What is software prototyping?
5. Write down the user interface design process.
6. What's design evaluation?
7. What is the need for software testing?
8. Write short notes on white box testing.
9. What is risk refinement?
10. What do you mean by statistical quality assurance?

PART B — (5 × 5 = 25 marks)

Answer ALL questions, Choosing either (a) or (b) in each.

11. (a) Explain the umbrella activities of software process.

Or

(b) List the task regions in spiral model.

12. (a) Discuss on Object oriented analysis of a software.

Or

(b) Explain the objectives of analysis modelling.

13. (a) Discuss on Data design activities.

Or

(b) List down the golden rules of user interface design.

14. (a) Explain activities in unit testing and integration testing.

Or

(b) Briefly explain the various software metrics.

15. (a) Discuss: Reactive vs Proactive risks.

Or

(b) Discuss the activities of Software Quality assurance.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Explain the following software process models :

    (a) Linear sequential model

    (b) RAD model.

17. Explain the activities of Requirement engineering.

18. Describe the various activities in architectural design of a software process.

19. Explain various testing strategies.

20. Explain Risk protection and risk refinement.

———————

D–6903

Sub. Code
31522

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Second Semester

Computer Applications

RELATIONAL DATABASE MANAGEMENT SYSTEMS (RDBMS)

(CBCS 2018-19 Aca.Year onwards & 2020-21 Aca.Year Onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Expand and write the purpose of the term DBMS.

2. Give the levels of data abstraction.

3. What is candidate key?

4. Write the use of rename operation.

5. What is trigger?

6. List out the desirable properties of decomposition.

7. What are the properties of transaction?

8. What are the types of serializabilty?

9. What are called secondary indices?

10. Define the term search key.

PART B — (5 × 5 = 25 marks)

Answer ALL questions, Choosing either (a) or (b).

11. (a) Discuss the role of DBA and different database end users.

Or

(b) Write short notes on various types of attributes in ER model.

12. (a) Discuss on tuple relational calculus

Or

(b) Give a note on domain relational calculus.

13. (a) Define BCNF. How does it differ from 3NF?

Or

(b) Brief on functional dependency concepts.

14. (a) Write short notes on concurrency.

Or

(b) Describe the timestamp based protocols.

15. (a) Give a note on comparison of file organizations.

Or

(b) Briefly explain about ISAM.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. With a neat diagram, explain the structure of DBMS.

17. Explain various integrity constraints giving examples.

18. Explain five SQL commands and their use, through examples.

19. Discuss on recovery and atomicity.

20. Explain B+ tree indexing.

D–6904

Sub. Code
31523

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Second Semester

Computer Applications

COMPUTER GRAPHICS

(CBCS 2018-19 Aca.Year onwards & 2020-21 Aca.Year onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. What is Computer graphics?

2. Write any two characteristics of video display devices.

3. Why we do rotation in computer graphics?

4. What is two dimensional computer graphics?

5. What is space partitioning representation in computer graphics?

6. What are the three different classification of curves?

7. What is called a shear in 3d Geometric transformation?

8. What are the three types of orthographic projection?

9. What are the two approaches to remove hidden surface problem?

10. What is the purpose of frame buffer method?

PART B — (5 × 5 = 25 marks)

Answer ALL the questions, Choosing either (a) or (b).

11. (a) Briefly discuss about cathode ray tube with a neat sketch.

Or

(b) Write short notes on flood fill algorithm.

12. (a) Elaborate on the translation operation with necessary diagrams and matrix.

Or

(b) What is text clipping? Explain with a neat diagram.

13. (a) Bring out the advantages and disadvantages of polygon meshes.

Or

(b) Write short notes on the following:

(i) Explicit curves,

(ii) Parametric curves.

14. (a) What is reflection? Explain with a neat structure.

Or

(b) What are the two basic projection methods? Explain.

15. (a) Explain briefly about the area sub division method with its diagram.

Or

(b) Write short notes on A- buffer method.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Write and explain in detail about the DDA line drawing algorithm.

17. Elaborate in detail the cohen-sutherland line clipping algorithm with a neat diagram.

18. List out the properties of B-Spline curves with a neat structure.

19. Discuss in detail about polygon clipping with a neat structure.

20. Describe in detail about the different types of animation techniques with illustrations.

———————————

D–6905

Sub. Code
31531/ 34031

DISTANCE EDUCATION

M.C.A./M.C.A.(Lateral Entry) DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Third Semester

Computer Application

DISCRETE MATHEMATICS

(CBCS – 2018-19 Academic Year Onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Write the following statement in symbolic form : "If either Jerry takes C++ Programming or Ken takes Java programming, then Larry will take Cyber Security".

2. Give the power set of $\{a, \{b\}\}$.

3. Give examples of sets A, B, C such that $A \cup B = A \cup C$ but $B \neq C$.

4. Give an example of a relation which is neither reflexive nor irreflexive.

5. Construct a function $f : \mathbb{N} \to \mathbb{N}$ such that f is 1–1 but not onto.

6. Show that the complement of the complement of A is itself.

7. Define Monoid with an example.

8. Let $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix}$. Find $\alpha\beta$.

9. Draw all possible simple digraphs having three nodes.

10. Write a sample space when 3 coins are tossed.

SECTION B — (5 × 5 = 25 marks)

Answer ALL questions, choosing either (a) or (b).

11. (a) Construct a truth table of $\neg(P \vee (Q \wedge R)) \rightleftharpoons ((P \wedge Q) \vee (P \wedge R))$.

Or

(b) Obtain conjunctive normal form of $\neg(P \vee Q) \rightleftharpoons (P \wedge Q)$.

12. (a) Let L denote the relation "less than or equal to" and D denote the relation "divides" when xDy means "x divides y". Both L and D are defined on the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Write L and D as sets and find $L \cap D$.

Or

(b) Let $X = \{1, 2, 3, 4\}$ and $R = \{\langle x, y\rangle \mid x > y\}$. Draw the graph of R and its matrix.

13. (a) If $f : X \to Y$ and $g : Y \to Z$ are onto, show that $g \circ f$ is also onto. Is $g \circ f$ is one-to-one if both g and f are one-to-one?

Or

(b) Show that $f(A \cup B) = f(A) \cup f(B)$ and $f(A \cap B) \subseteq f(A) \cap f(B)$. Under what condition will $f(A \cap B) = f(A) \cap f(B)$?

14. (a) Show that the intersection of any two congruence relations on a set is also a congruence relation.

Or

(b) Show that the set of all the invertible elements of a monoid form a group under the same operation as that of the monoid.

15. (a) Show that, in a simple digraph with n nodes, the length of any elementary cycle does not exceed n.

Or

(b) A random variable X is defined as the sum of the numbers on the faces when two dice are thrown. Find the expected values of X.

SECTION C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Show that the following premises are inconsistent :

(a) If Jack misses many classes through illness, then he fails high school.

(b) If Jack fails high school, then he is uneducated.

(c) If Jack reads a lot of books, then he is not uneducated.

(d) Jack misses many classes through illness and reads a lot of books.

17. (a) Define composition of binary relations.

(b) Let $R = \{\langle 1, 2\rangle, \langle 3, 4\rangle, \langle 2, 2\rangle\}$ and $S = \{\langle 4, 2\rangle, \langle 2, 5\rangle, \langle 3, 1\rangle, \langle 1, 3\rangle\}$. Find $R \circ S$, $S \circ R$, $R \circ (S \circ R)$, $(R \circ S) \circ S$, $R \circ R$, $S \circ S$, $R \circ R \circ R$ and $S \circ S \circ S$.

18. Let F_x be set of all one-to-one onto mappings from X onto X, where $X = \{1, 2, 3\}$. Find all the elements of F_x and find the inverse of each element.

19. State and prove the fundamental theorem of group homomorphism.

20. The probability that India wins a cricket test match against Pakistan is know to be 2/5. If India and Pakistan play 3 test matches what is the probability that :

(a) India will win all the test matches,

(b) India win atleast one test match,

(c) India will win atmost one match,

(d) India will loose all the three matches.

D–6906

Sub. Code
31532/34032

DISTANCE EDUCATION

M.C.A/M.C.A (Lateral entry) DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Third Semester

Computer Application

OPERATING SYSTEM

(CBCS 2018 – 2019 Academic Year Onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. What is meant by an Operating System?

2. What are system calls?

3. What is meant by process scheduling?

4. Define the term scheduling.

5. What is semaphore?

6. How to detect a dead lock?

7. What is swapping?

8. Define the term paging.

9. What are file access methods?

10. How do you protect a file?

PART B — (5 × 5 = 25 marks)

Answer ALL questions choosing either (a) or (b).

11. (a) What are the services provided by Operating System? Explain.

Or

(b) Write short notes on structure of the Operating System.

12. (a) Discuss on inter process communication.

Or

(b) Write a detailed note on multiple processor scheduling.

13. (a) Explain the critical section problem.

Or

(b) How do you perform recovery from deadlock? Explain

14. (a) Describe about contiguous memory allocation.

Or

(b) Write brief notes on memory segmentation.

15. (a) Illustrate on file system mounting.

Or

(b) Describe about disk scheduling and management.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Explain the Operating System design and implementation.

17. Discuss any two scheduling algorithms.

18. Describe the methods for handling deadlocks.

19. Briefly explain the memory management strategies.

20. Discuss about implementing file systems.

———————

D–6907

Sub. Code
31533/34033

DISTANCE EDUCATION

M.C.A/M.C.A (Lateral entry) DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Third Semester

Computer Application

OBJECT ORIENTED ANALYSIS AND DESIGN

(CBCS 2018 – 2019 Academic Year Onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. What are the orthogonal views of software?

2. Why is encapsulation important?

3. What is the main advantage of DFD?

4. What is data modeling?

5. Mention some of the forms of association.

6. How will you identify actors?

7. Write the guidelines for naming classes.

8. What is white box testing?

9. State the design axioms.

10. What is client/server computing?

PART B — (5 × 5 = 25 marks)

Answer ALL questions, choosing either (a) or (b).

11. (a) Describe the components of unified approach.

Or

(b) Give a note on advantages and disadvantages of prototyping.

12. (a) What is an object model? Explain other OMT models.

Or

(b) Write a note on Frameworks.

13. (a) Describe the basic activities in object oriented analysis.

Or

(b) How Classes, Responsibilities and Collaborators are useful?

14. (a) List and explain object-oriented design corollaries.

Or

(b) Brief on database models.

15. (a) Write the guidelines for designing Forms and Data Entry Windows.

Or

(b) Discuss on Foundation class library.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Explain object oriented system development.

17. Describe the Booch system development process.

18. Explain the approaches for identifying classes.

19. Explain the various testing strategies.

20. Create user satisfaction test for a bank application

———————

D-6908

Sub. Code
31541/34041

DISTANCE EDUCATION

M.C.A./ M.C.A. (Lateral Entry) DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Fourth Semester

Computer Application

ACCOUNTING AND FINANCIAL MANAGEMENT

(CBCS 2018–2019 academic year onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal mark s.

1. What is book keeping?
2. State the uses of trial balance.
3. What is P/V ratio?
4. Give the meaning of prime cost.
5. What are material variances?
6. What is master budget?
7. Explain the concepts discounting and compounding.
8. What are the different types of working capital?
9. What do you mean by the term ‘capital structure’?
10. What is stable dividend policy?

PART B — (5 × 5 = 25 marks)

Answer ALL the questions Choosing either (a) or (b).

All questions carry equal marks.

11. (a) During the year 2009, Soni Ltd., earned a profit of Rs. 1,85,720 after adjusting the following :

	Rs.
Provision for bad debts	1,500
Salaries	8,500
Depreciation written off	15,300
Profit on sale of fixed assets	14,000
Discount on debentures written off	20,000
Loss on sale of investments	2,000
Preliminary expenses written off	8,000
Proposed dividend	50,000
Transfer to debenture redemption fund	20,000
Dividend received	4,500

Calculate funds from operations.

Or

(b) Explain the salient features of Tally accounting software.

12. (a) From the following data calculate :

(i) P/V Ratio and

(ii) Profit

Sales	Rs. 20,000
Fixed Expenses	Rs. 4,000
Break Even Point	Rs. 10,000

Or

(b) 'Cost Volume Profit Analysis is helpful for profit planning' — Explain.

13. (a) For making 10 kg of yarn, the standard material requirement is :

Material	Quantity (kg.)	Rate per kg. (Rs.)
White	8	6.00
Black	4	4.00

In March, 1000 Kg of yarn was produced. The actual consumption of material is as under :

Material	Quantity (kg.)	Rate per kg. (Rs.)
White	750	7.00
Black	500	5.00

Calculate :

(i) Material Cost Variance

(ii) Material Price

(iii) Material Usage Variance.

Or

(b) What is Zero Base Budgeting? What are its advantages?

14. (a) 'The goal of financial decision making should be Wealth Maximisation rather than Profit Maximisation" — Comment.

Or

(b) Raman & Co., is considering the purchase of a machine. Two machines A and B each costing Rs.50,000 are available. Cash inflows expected to be as under. Calculate the Pay- back Period.

Year	Cash Inflows	
	Project A	Project B
1	15,000	5,000
2	20,000	15,000
3	25,000	20,000
4	15,000	30,000
5	10,000	20,000

15. (a) Explain the various financial leverages.

Or

(b) Examine the Modigliani and Miller Approach of dividend policy.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

All questions carry equal marks.

16. Following are the balances extracted from the books of Mr. Natarajan as on 31st March 2018.

Particulars	Debit (Rs.)	Credit (Rs.)
Purchases and Sales	10,000	15,100
Wages	600	
Freight inwards	750	
Advertisement	500	
Carriage outwards	400	
Commission received		1,900
Rent received		600
Machinery	8,000	
Debtors and Creditors	2,250	2,400
Bills receivable	300	
Cash	1,200	
Stock on 1st Jan. 2018	1,000	
Capital		5,000
	25,000	25,000

Additional information :

(a) Commission received in advance Rs.400

(b) Advertisement paid in advance Rs. 150

(c) Wages outstanding Rs.200

(d) Closing stock on 31st March 2018, Rs.2,100

Prepare Trading account, Profit & Loss Account and Balance Sheet.

17. During the year 2008, XYZ Ltd., produced 50,000 units of a product. The following were the expenses :

	Rs.
Stock of raw materials on 1.1.2008	10,000
Stock of raw materials on 31.12.2008	20,000
Purchases	1,60,000
Direct wages	75,000
Direct expenses	25,000
Factory expenses	37,500
Office expenses	62,500
Selling expenses	25,000

You are required to prepare a Cost Sheet showing cost per unit and total cost at each stage.

18. Prepare a Cash Budget for the months of May, June and July 2001 on the basis of the following information :

(a) Income and Expenditure forecasts:

Month 2001	Credit Sales	Credit Purchases	Wages	Mfg. Expenses	Office Expenses	Selling Expenses
	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.
March	60,000	36,000	9,000	4,000	2,000	4,000
April	62,000	38,000	8,000	3,000	1,500	5,000

Month 2001	Credit Sales	Credit Purchases	Wages	Mfg. Expenses	Office Expenses	Selling Expenses
	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.
May	64,000	33,000	10,000	4,500	2,500	4,500
June	58,000	35,000	8,500	3,500	2,000	3,500
July	56,000	39,000	9,500	4,000	1,000	4,500
August	60,000	34,000	8,000	3,000	1,500	4,000

(b) Cash balance on 1st May 2001 Rs.8,000.

(c) Plant costing Rs. 16,000 is due for delivery in July; payable 10% on delivery and the balance after 3 months.

(d) Advance tax of Rs.8,000 each is payable in March and June.

(e) Period of credit allowed

(i) by suppliers – 2 months and

(ii) to customers – 1 month

(f) Lag in payment of manufacturing expenses – half month

(g) Lag in payment of office and selling expenses – 1 month.

19. Discuss the factors influencing working capital requirements of a company.

20. Briefly explain the factors affecting dividend policy decisions.

D– 6909

Sub. Code
31542/34042

DISTANCE EDUCATION

M.C.A/M.C.A. (Lateral Entry) DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Fourth Semester

COMMUNICATION SKILLS

(CBCS-2018 - 19 ACADEMIC YEAR ONWARDS)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Define the term Communication.

2. What are the different forms of communication?

3. Why are soft skills important?

4. How do you disagree with your higher official? Write a set of two expressions.

5. How is oral presentation different from oral communication?

6. Why should the presenter analyse the occasion for presentation?

7. What is Group Discussion?

8. Why is it important to have meetings?

9. What are the key components of a job application letter?

10. State the importance of Writing agenda.

PART B — (5 × 5 = 25 marks)

Answer ALL questions, Choosing either (a) or (b).

11. (a) Bring out the importance of communication.

Or

(b) Enumerate the major barriers to effective communication.

12. (a) Distinguish between verbal and non–verbal communication.

Or

(b) Write any five conversational sentences for making request.

13. (a) Write a note on audience analysis.

Or

(b) Explain the term chronemics with examples.

14. (a) Discuss briefly on the participation techniques in-group communication.

Or

(b) Analyse the objectives of conducting a meeting.

15. (a) What are the advantages of e–mail?

Or

(b) Write a note on formal and non–formal reports.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Analyse the channels of communication.

17. Explain the importance and attributes of soft skills.

18. Discuss the essential elements in the introduction, middle and conclusion of an oral presentation.

19. Comment on the effective skills needed for teamwork.

20. Write a detailed resume along with a covering letter applying for the post of manager in a multinational company.

D–6910

Sub. Code
31543/34043

DISTANCE EDUCATION

M.C. A/ M.C.A (Lateral entry) DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

FOURTH SEMESTER

Computer Application

INTERNET AND JAVA PROGRAMMING

(CBCS 2018 – 2019 Academic Year Onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. What is the purpose of FTP? Write down the default port number allocated for FTP.

2. Write down any four Internet search engine providers.

3. What is meant by token in Java?

4. What is called JVM?

5. Define the term interface in java.

6. What is called final variable in java?

7. Write down any four method names in java thread class.

8. What is meant by runnable interface.

9. What is called character stream in java?

10. Write down syntax to read and write files in java.

PART B — (5 × 5 = 25 marks)

Answer ALL questions choosing either (a) or (b).

11. (a) Give a brief account of SMTP

Or

(b) Write short notes on messaging in internet.

12. (a) What is called labeled loop? Explain with suitable example

Or

(b) Write short notes on Type casting in Java.

13. (a) How to access class members in java? Explain with suitable example

Or

(b) Explain about overloading methods.

14. (a) Write a java program to draw a rectangle and circle using graphic class and methods

Or

(b) Describe the process of Thread synchronization.

15. (a) Write short notes on I/O exception class.

Or

(b) Write a java program to read and write characters using FileReader and FileWriter.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Explain in detail about features of Email.

17. Explain about decision making and branching in Java with suitable examples.

18. How to implement interface and package? Explain with program code.

19. Discuss about Applet life cycle with suitable example.

20. Discuss in detail about byte stream classes.

———————

D–6911

Sub. Code
31551/34051

DISTANCE EDUCATION

M.C. A/ M.C.A (Lateral) DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Fifth Semester

Computer Applications

COMPUTER NETWORKS

(CBCS 2018 – 2019 Academic Year Onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Write any two applications of Computer networks.

2. Differentiate between LAN and MAN.

3. What is CSMA/CD?

4. What is called selective repeat?

5. What is the advantage of packet switching?

6. Differentiate between static routing and dynamic routing.

7. Define the term virtual circuit.

8. What is UDP?

9. What do you mean by DNS?

10. What is WWW?

PART B — (5 × 5 = 25 marks)

Answer ALL questions choosing either (a) or (b).

11. (a) Briefly explain the various network topologies.

Or

(b) Compare and contrast: Analog vs Digital signal performance.

12. (a) Write about cyclic redundancy check.

Or

(b) Discuss on flow and error control.

13. (a) Explain ARQ.

Or

(b) What do you mean by Datagram subnet? Explain.

14. (a) Explain Hierarchical routing.

Or

(b) Write short notes on Flooding.

15. (a) Compare and contrast: Connection oriented vs Connectionless service.

Or

(b) Write short notes on SNMP

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Describe the various layers of OSI reference model.

17. Explain the various Transmission media in detail.

18. Explain Error detection and correction techniques.

19. Explain the Dynamic routing algorithms.

20. Discuss on:

    (a) File transfer

    (b) Remote procedure call.

———————

D–6912

Sub. Code
31552/34052

DISTANCE EDUCATION

M.C. A/ M.C.A (Lateral Entry) DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Fifth Semester

DATA MINING AND WAREHOUSING

(CBCS 2018 – 2019 Academic Year Onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. What is called OLAP?

2. What is meant by Data Enrichment?

3. Define the term Unsupervised Learning.

4. What is meant by Support and Confident in Rule Mining?

5. Write the various applications of Clustering.

6. Define Hierarchical Clustering?

7. Write the data mining features available in Rapid minerTool?

8. What is called spatial mining?

9. Write the difference between Traditional and Big Data Approaches?

10. What are the limitations of Hadoop?

PART B — (5 × 5 = 25 marks)

Answer ALL questions choosing either (a) or (b).

11. (a) Write short notes on Snow flake Schema in Data warehousing.

Or

(b) Exemplify about the Current Trends in Data Mining.

12. (a) Elucidate the Applications of Association Rule Mining.

Or

(b) Explain briefly about Bayesian Classification Algorithm.

13. (a) Explain briefly about DBSCAN Algorithm.

Or

(b) Differentiate between machine learning and data mining.

14. (a) Write about the Techniques involved in Text Clustering.

Or

(b) Write about Techniques involved in temporal mining.

15. (a) What are the characteristics of Big Data?

Or

(b) What are the different types of data come under Big Data?

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Explain in Detail about the Architecture of Data warehouse.

17. Explain in detail about Decision Tree Induction Algorithm.

18. Explain in detail about:

(a) Web Content Mining.

(b) Web Usage Mining.

19. Explain pincher search Algorithm.

20. Explain about temporal Mining.

———————

D–6913

Sub. Code
31553/34053

DISTANCE EDUCATION

M.C. A/ M.C.A (Lateral) DEGREE EXAMINATION.

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Fifth Semester

Computer Applications

VISUAL PROGRAMMING WITH .NET

(CBCS 2018 – 2019 Academic Year Onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. What is visual studio?

2. Differentiate between Windows project and Web project.

3. What is the use of `enum' data type?

4. What is Property snippet?

5. What is Array? List any two benefits of it.

6. What is the use of class designer?

7. Write about the auto window.

8. What is called as foreign key? List the uses.

9. Write about WPF.

10. What is MVC? List its advantages.

PART B — (5 × 5 = 25 marks)

Answer ALL questions choosing either (a) or (b).

11. (a) Explain the following Visual Studio components: The Menu, Toolbox and Work Area.

Or

(b) Explain about Visual Studio Windows management.

12. (a) Write short notes on Code Skeleton.

Or

(b) Write a program to display names of 5 cities using abstract class inheritance.

13. (a) Explain Arrays an Generics.

Or

(b) Explain about compilation of a Project.

14. (a) Discuss on Application state inspection.

Or

(b) Write the procedure to create an application in Visual Studio for student data management.

15. (a) Detail on Layouts.

Or

(b) Explain the procedures to use WPF controls.

PART C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Explain the Project types in Visual Studio.

17. Discuss in detail about VB and C# Loop controls.

18. How do you examine the property settings? Explain.

19. `Working with data in Visual Studio' – Elaborate.

20. Write detailed notes on the design of Silverlight Applications.

———————

D–7338

Sub. Code
31561/34061

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral Entry) DEGREE EXAMINATION

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Sixth Semester

Computer Application

CLOUD COMPUTING

(CBCS 2018 – 2019 Academic Year Onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. What is cloud computing?

2. Mention the benefits of cloud computing.

3. What are the various cloud services?

4. List out the various types of tools used is cloud service development.

5. Define the term data leakage.

6. Give the advantages of Force.com for Google App engine.

7. What is Cast Iron Cloud?

8. List the popular cloud applications.

9. List the different levels of migration.

10. Name the four levels of federation in cloud.

SECTION B — (5 × 5 = 25 marks)

Answer ALL the questions, choosing either (a) or (b).

11. (a) Discuss the pros and cons of cloud computing.

Or

(b) List out system issues for running typical parallel program in cloud data centers.

12. (a) State and explain service models of cloud computing.

Or

(b) Explain SQL Azure and Azure tables.

13. (a) Discuss cloud computing for corporation.

Or

(b) What is Amazon S3? Explain in detail.

14. (a) How will you collaborate word processing on cloud? Explain.

Or

(b) How will you store and share files in cloud architecture?

15. (a) Draw two level architecture of resource allocation in cloud.

Or

(b) Discuss the privacy issues in cloud.

SECTION C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Describe the different system models for distributed and cloud computing.

17. Compare and contrast physical versus virtual clusters.

18. Discuss the various cloud services or applications available to individuals.

19. Discuss in detail about cloud networks.

20. Illustrate the various issues in cloud computing.

———————

D–7339

Sub. Code
31562/34062

DISTANCE EDUCATION

MCA/MCA (Lateral) DEGREE EXAMINATION

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Sixth Semester

SOFT COMPUTING

(CBCS 2018 – 2019 Academic Year Onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Define Hebb's network.

2. Define Axon.

3. Define the term Perceptron.

4. What is meant by RBF?

5. List the application of Fuzzy logic.

6. Define Defuzzification.

7. What is Fuzzy Inference system?

8. What is the need for Fuzzy rule?

9. What is meant by selection?

10. Write a note on Inversion.

SECTION B — ($5 \times 5 = 25$ marks)

Answer ALL the questions, choosing either (a) or (b).

11. (a) Explain about Linear separability.

Or

(b) Differentiate Soft Vs Hard computing.

12. (a) Explain about Madaline networks.

Or

(b) Write a note on ART.

13. (a) Write a note on Fuzzy equivalence.

Or

(b) Write a note on Fuzzy composition.

14. (a) Explain about expert system and its applications.

Or

(b) Discuss the formation and decompositions of rules.

15. (a) Explain about Crossover.

Or

(b) Write a note on genetic programming.

SECTION C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Briefly explain about ANN architecture.

17. Briefly explain about Boltzmann machine.

18. Discuss types methods of membership value assignments.

19. Explain in detail about Fuzzy propositions.

20. Explain GA Algorithm and its functionalities.

D–7340

Sub. Code
31563/34063

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral Entry) DEGREE EXAMINATION

MAY 2021 EXAMINATION

&

MAY 2020 ARREAR EXAMINATION

Sixth Semester

Computer Application

BIG DATA ANALYTICS

(CBCS 2018 – 2019 Academic Year Onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

1. List the characteristics of big data.

2. Write the limitations of Hadoop.

3. List the algorithms that use map reduce.

4. What are similar items?

5. What is meant by filtering a stream?

6. What is meant by decaying window?

7. What is the job of search engine?

8. Define the term Hub.

9. What is the purpose of recommendation system?

10. Name the applications of social network mining.

SECTION B — (5 × 5 = 25 marks)

Answer ALL questions, choosing either (a) or (b).

11. (a) Describe physical architecture of Hadoop.

Or

(b) Briefly explain about the technologies available for Big data analytics.

12. (a) Write short notes of similarity of documents.

Or

(b) Write short notes on software stack.

13. (a) What is stream query? Explain briefly.

Or

(b) Give a brief account on sampling in data streams.

14. (a) Brief on the history of search engines.

Or

(b) Explain briefly about Topic-Sensitive page rank.

15. (a) What is the use of recommendation system? Give an example.

Or

(b) Write short notes on mining social network graph.

SECTION C — (3 × 10 = 30 marks)

Answer any THREE questions.

16. Describe core components of Hadoop.

17. Explain in detail about nearest neighbor search.

18. Explain in detail about Querying on Window.

19. Describe Hubs and authorities with reference to page ranking.

20. Explain about counting triangles in social graph.

———————